

TRANSPORT AND THE SUPERCRITICALITY GAP IN 3D NAVIER–STOKES, IV: EVENTS

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ABSTRACT. Part III stated the missing estimate for the three-dimensional Navier–Stokes equations as a question about chains of strain-producing events: whether the escalation of intensity per halving of scale stays below the critical value 2. There the events were defined in simulations. Here we define them rigorously and follow the question as far as it goes. In Lagrangian variables an event is a unit of pair stretching of the stochastic flow. Each unit costs a unit of $\int \|\nabla u\|_{L^\infty} dt$, a singularity is infinitely many units in finite time, and the correct event scale is the current extent of the stretched pairs, not the size of the stretched set. In frequency space, a band at the viscous edge must, given only the energy, concentrate on a set of effective dimension at most one, while concentration above dimension one gives regularity through the criterion of Yoneda, Goto and Tsuruhashi; the two formalizations meet at one critical line. The currency of the ledger is circulation: by the stochastic Kelvin theorem of Constantin and Iyer it is never created, only compressed from larger scales and lost to the noise average, and every known blowup construction for a neighbouring equation lacks either a circulation theorem or the average. In the exact reduction of a vortex tube in an external linear strain we prove a sharp viscous threshold: the mean-square radius cannot fall below $2\nu/\gamma$, a structure cannot compress itself, and only axial stretching compresses, which also accounts for curvature. Registered simulations with identified tubes find them at this strain radius, compressed mainly by the collective background rather than by distinct stretchers. An event dissipation budget and the circulation picture then place a critical chain at the classical frontier: an active set of dimension below one, with velocity of order ν/r at scale r . Finally we give the anatomy of a collapse at any rate. The reduction permits every rate; a Navier–Stokes realization needs suppliers of total circulation Reynolds number at least 8π within a few core radii; and a Type II collapse must either shrink below the parabolic scale, where it rescales to a bounded eternal solution, or gather circulation.

1. INTRODUCTION

Part III [27] introduced hand-off chains of producing events, with intensities I_k , scales ℓ_k and event Reynolds numbers $R_k = I_k \ell_k^2 / \nu$, and proved the bookkeeping identity $\log(R_n/R_0) = \sum_{k < n} (\alpha_k - 2) w_k$. Under an *event picture* taken from simulations, a singularity requires chains whose mean escalation reaches 2, and the missing estimate becomes the question whether it stays below $2 - \delta$ [27, Question 6.3]. This part replaces the event picture by definitions and pushes the question to its end.

Sections 2–4 define events in Lagrangian variables and in frequency space and show that both lead to the same critical line; the missing ingredient in both is a viscous threshold. Section 5 identifies circulation as the conserved currency of the ledger. Sections 6–8 prove the threshold in the exact reduction of a stretched vortex tube, first for Burgers vortices, then for any profile and any planar strain, and show that only axial stretching compresses; registered simulations test each statement on identified structures. Section 9 shows that the question, pushed to its end, lands at the classical frontier, and Section 10 applies the tools to collapse at any rate: the reduction permits every rate, a Navier–Stokes realization needs a cluster of vortices stretching

one another, and the Type I case and the two kinds of Type II collapse land at different classical questions.

What is new, and what is not. The stochastic Lagrangian representation and its Kelvin theorem are due to Constantin and Iyer [1]; the stretching functional to Crippa and De Lellis [2]; the triad criterion to Yoneda, Goto and Tsuruhashi [24]; the observation that common noise cancels in separations to Mahithitarmmatorn [23]. The second-moment identity behind Theorem 7.1 is classical in the stability theory of Burgers vortices [7], and a stronger form of the concentration monotonicity behind Theorem 8.1 is proved for bounded divergence-free drifts in [19]. The measured balance of tubes at their strain radius replicates the finding of Jiménez and Wray [20]. The dimension-one threshold and the collapse scenarios of §§9–10 are the frontier set by Caffarelli, Kohn and Nirenberg [4], by the self-similar theory [9, 11, 12], and by the Liouville theory of Koch, Nadirashvili, Seregin and Šverák [13]. What we add is: the rigorous level chains and the correction of the naive event scale (Remark 2.8); the edge proposition and the triad–concentration lemma (Proposition 3.3, Lemma 3.4); the circulation formula and capacity bound (Proposition 5.1, Corollary 5.2); the use of the second moment and of concentration monotonicity with an external strain to obtain the threshold (Theorems 7.1 and 8.1); the event dissipation budget (Proposition 9.1); the rate-independent supplier bound and the splitting of Type II collapse (Corollary 10.2, Remark 10.3); and the registered measurements, including their failures.

2. EVENTS IN LAGRANGIAN VARIABLES

Let u be a smooth solution on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ on its maximal interval $[0, T^*)$. For $s \leq t < T^*$ let $X_{s,t}$ be the stochastic flow of Constantin and Iyer [1] from time s ,

$$X_{s,t}(a) = a + \int_s^t u(X_{s,r}(a), r) dr + \sqrt{2\nu} (W_t - W_s),$$

so that $X_{s,t} = X_{r,t} \circ X_{s,r}$ and, pathwise, $\nabla X_{s,t}$ satisfies $\frac{d}{dt} \nabla X_{s,t} = (\nabla u)(X_{s,t}, t) \nabla X_{s,t}$. Hence $\text{Lip}(X_{s,t})$ and $\text{Lip}(X_{s,t}^{-1})$ are at most $\exp \int_s^t \|\nabla u\|_{L^\infty}$. Distances between points are computed on lifts to $\mathbb{R}^3$.

Lemma 2.1 (common noise). $X_{s,t}(a) - X_{s,t}(b) = a - b + \int_s^t (u(X_{s,r}(a), r) - u(X_{s,r}(b), r)) dr$.

The noise does not depend on the label and cancels in every separation (compare [23, Prop. 4.2]). Everything below is built from separations, so the noise enters only through the points at which u is sampled.

The stretching functional. Fix $R_0 = \pi/2$. For a map Y of $\mathbb{T}^3$, $x \in \mathbb{T}^3$ and $0 < r \leq R_0$, let

$$Q_r(Y)(x) = \frac{1}{|B_r|} \int_{B_r(x)} \log \left(1 + \frac{|Y(x) - Y(y)|}{r} \right) dy, \quad Q^*(Y) = \sup_{0 < r \leq R_0} Q_r(Y).$$

This is the functional of Crippa and De Lellis [2], used in Part II [26, §5]. For the identity, $Q_r(\text{id}) \equiv q_{\text{id}} = \frac{1}{|B_1|} \int_{B_1} \log(1 + |z|) dz = 2 \log 2 - \frac{5}{6} \approx 0.553$. For smooth Y , $Q_r(Y)(x)$ extends continuously to $r = 0$, and

$$Q^*(Y) \leq \log(1 + \text{Lip } Y), \quad \text{Lip } Y \leq e^{C(1+2\|Q^*(Y)\|_{L^\infty})} \quad (1)$$

with an absolute constant C [26, §5].

Lemma 2.2 (composition). *For smooth maps Y, Z and all r, x : $Q_r(Z \circ Y)(x) \leq Q_r(Y)(x) + \log^+ \text{Lip } Z$.*

Proof. $|Z(Y(x)) - Z(Y(y))| \leq \text{Lip } Z |Y(x) - Y(y)|$, and $\log(1 + La) \leq \log L + \log(1 + a)$ for $L \geq 1$, $a \geq 0$. $\square$

Corollary 2.3. *For $s \leq t' \leq t < T^*$ and every realization, $|Q^*(X_{s,t}) - Q^*(X_{s,t'})| \leq \int_{t'}^t \|\nabla u\|_{L^\infty}$. Consequently the level function $f_s(t) = \text{ess sup}_W \|Q^*(X_{s,t})\|_{L^\infty(\mathbb{T}^3)}$ is continuous, $f_s(s) = q_{\text{id}}$, and $|f_s(t) - f_s(t')| \leq \int_{t'}^t \|\nabla u\|_{L^\infty}$.*

Proof. Apply Lemma 2.2 to $X_{s,t} = X_{t',t} \circ X_{s,t'}$ and to $X_{s,t'} = X_{t',t}^{-1} \circ X_{s,t}$. $\square$

Levels.

Definition 2.4. For an integer $n \geq 1$, the *level time* is $t_n(s) = \inf\{t \in [s, T^*) : f_s(t) \geq n\}$, with $\inf \emptyset = T^*$. The climb from level n to level $n+1$ is the interval $[t_n, t_{n+1})$, and its *rate* is $I_n = 1/(t_{n+1} - t_n)$.

Proposition 2.5 (each level costs a unit of strain-time). *If $t_m(s) < T^*$ and $n < m$, then $\int_{t_n}^{t_m} \|\nabla u\|_{L^\infty} dt \geq m - n$. In particular the mean of $\|\nabla u\|_{L^\infty}$ over a climb is at least its rate I_n .*

Proof. By continuity $f_s(t_n) = n$ and $f_s(t_m) = m$; apply Corollary 2.3. $\square$

Proposition 2.6 (singularity means infinitely many levels). *Let $s < T < \infty$. The solution is smooth beyond T if and only if f_s is bounded on $[s, \min(T, T^*))$. If $T^* < \infty$, then $t_n(s) < T^*$ for every n and $t_n(s) \uparrow T^*$; in particular $\sum_n 1/I_n < \infty$ and $I_n \rightarrow \infty$.*

Proof. By (1), f_s is bounded if and only if $\text{Lip } X_{s,t}$ is bounded uniformly in t and in the noise, which by [26, Thm. 3.1], applied to the solution started at time s , is equivalent to smoothness beyond T . If $T^* < \infty$, f_s is unbounded on $[s, T^*)$, so every level is reached; t_n is increasing and cannot converge to some $t_\infty < T^*$, since f_s is bounded on $[s, t_\infty]$. Finally $\sum_n (t_{n+1} - t_n) \leq T^* - s$, so $t_{n+1} - t_n \rightarrow 0$. $\square$

Proposition 2.7 (a priori size of level sets [2, 26]). *For $n > 2 \log 2$ and every realization, the set $\{Q^*(X_{s,t}) > n\}$ has measure at most $4C^4(t-s)D_{[s,t]}/(\nu n^2)$, where $D_{[s,t]} = \nu \int_s^t \|\nabla u\|_{L^2}^2$.*

The event scale. The natural first guess for the scale of the level- n event is the size of the set of labels stretched past level n . It is wrong.

Remark 2.8 (the size of the stretched set is not the event scale). Let u be smooth and steady near a hyperbolic stagnation point with strain rate γ in a region of size L . We sketch the behaviour. A label has $Q^* \geq n$ at time t only if its trajectory stays within the hyperbolic region for a time of order n/γ , which confines it to a layer of width of order Le^{-n} about the stable manifold. So the set stretched past level n has measure of order $L^3 e^{-n}$, while the climbs keep a bounded duration of order $1/\gamma$. An event Reynolds number built from the size of the stretched set tends to zero exponentially in n although nothing singular happens. The size of the stretched set measures the *labels* that have been stretched. The quantity that should be compared with the viscous scale is the *current extent* of the structure that stretches them, which here stays of order L .

Definition 2.9. For a smooth map Y and a point x with $Q^*(Y)(x) > n$, the set of $r \in (0, R_0]$ with $Q_r(Y)(x) \geq n$ is nonempty, by continuity of Q_r in r up to $r = 0$, and the *coherence radius* is $\rho_n(Y)(x) = \sup\{r \in (0, R_0] : Q_r(Y)(x) \geq n\}$, and the *current extent* is

$$L_n(Y)(x) = \frac{1}{|B_\rho|} \int_{B_\rho(x)} |Y(x) - Y(y)| dy, \quad \rho = \rho_n(Y)(x).$$

Lemma 2.10. $L_n(Y)(x) \geq (e^n - 1) \rho_n(Y)(x)$.

Proof. By Jensen's inequality for the concave function $\log(1 + \cdot)$, $\log(1 + L_n/\rho) \geq Q_\rho(Y)(x) \geq n$. $\square$

At the hyperbolic point, ρ_n is of order Le^{-n} and L_n of order L : the coherently stretched ball has been drawn out to the size of the structure that stretched it. For the level- n event of the flow from s we set

$$L_n = \sup \left\{ L_n(X_{s,t_{n+1}})(x) : Q^*(X_{s,t_{n+1}})(x) > n \right\}, \quad R_n = \frac{I_n L_n^2}{\nu},$$

the supremum being taken also over realizations of the noise. With these definitions the bookkeeping identity of Part III holds exactly:

$$\log \frac{R_m}{R_n} = \sum_{k=n}^{m-1} \left(\log \frac{I_{k+1}}{I_k} - 2 \log \frac{L_k}{L_{k+1}} \right), \quad (2)$$

so that, along any stretch of the chain on which L_k decreases, writing the k -th term as $(\alpha_k - 2)w_k$ with $w_k = \log(L_k/L_{k+1}) > 0$, Corollary 6.2 of [27] applies verbatim to the level chain. The set in the supremum is nonempty, since $f_s(t_{n+1}) = n + 1$.

What remains. Propositions 2.5–2.7 and (2) are unconditional. What Part III took from the event picture, and what is still missing, is a viscous threshold.

Question 2.11 (Lagrangian viscous threshold). *Is there $R_c > 0$, depending only on scale-invariant quantities, such that for every smooth solution of the Navier–Stokes equations and every $s < T^*$, the level chain of the flow from s satisfies $R_n \geq R_c$ for all sufficiently large n ?*

Four comments.

- (1) A positive answer, together with $L_n \rightarrow 0$, makes the conclusion of Part III rigorous: a singular solution has a level chain whose mean escalation tends to the critical value 2 (Proposition 2.6, (2) and [27, Cor. 6.2]). Question 6.3 of [27] then becomes a precise regularity criterion.
- (2) The threshold is consistent with smooth flows: at the hyperbolic point of Remark 2.8, R_n stays of order $\gamma L^2/\nu$.
- (3) It cannot come from the noise acting on separations, which cancels (Lemma 2.1). It must come from the velocity field itself: from the fact that u is the average of the Weber field transported by the same flow, that is, from the self-consistency of the ledger [27, Prop. 2.1]. This is where the open problem of Mahithitarmmatorn on self-consistent covariance closure [23, §8(ii)] sits in the present language.
- (4) It is a different question from [26, Question 6.1]. That question bounds the measure of the stretched label sets, which Remark 2.8 shows can shrink exponentially in smooth flows. Question 2.11 bounds the current extent of the stretching structures against the viscous scale of the climb.

Toward the threshold. We record what can and cannot deliver Question 2.11.

Not the noise (Lemma 2.1).

Not an a priori dissipative scale. Viscosity makes u real-analytic, and a lower bound on the radius of analyticity $\delta(t)$ would bound from below the scale of every stretching structure. But a lower bound $\delta(t) \geq \varphi > 0$ holding for all $t < T^*$ with φ depending only on a priori quantities, in the usual Gevrey sense in which the radius comes with a bound on the corresponding Gevrey norm, would by itself give regularity, so every such bound for three-dimensional Navier–Stokes solutions must assume control that the energy does not supply [21, 17, 16]. For the Euler

equations the radius obeys $\delta(t) \geq C \exp(-C \int_0^t \|\nabla u\|_{L^\infty})$ [22]: it shrinks like e^{-n} after n levels, exactly as the coherence radius does at a hyperbolic point.

Not the bookkeeping alone. By (2), a stretch of the chain on which R_k is constant is exactly a stretch on which $\alpha_k = 2$. A chain held at the viscous edge, $R_k \approx R_c$, escalates at precisely the critical rate. So Question 6.3 of [27], with the rigorous events of this section, is equivalent to the statement that no level chain can stay above the viscous threshold while its extent shrinks to zero: every cascade eventually becomes sub-viscous. This is a precise form of the regularity problem, not a reduction of it.

What is left is the ledger. The strain of a child event is paid by its sources [27, Prop. 2.1], and a source shell can pay only so much. From the kernel [27, (1)], a single source at separation z induces strain of magnitude at most $\frac{3\sqrt{2}}{8\pi}|\omega||z|^{-3}$. Since the strain kernel has zero mean on spheres, subtracting at each radius the spherical mean $\bar{\omega}_r(x)$ of ω on $\partial B_r(x)$ gives

$$|S^{[a,b]}(x)| \leq \frac{3\sqrt{2}}{2} \int_a^b \sup_{|z|=r} |\omega(x-z) - \bar{\omega}_r(x)| \frac{dr}{r}. \quad (3)$$

A shell pays in proportion to the oscillation of the vorticity across it, with an absolute constant per octave of source distance. In a chain with a halving of scale per step, $\alpha_k = 2$ requires the intensity to quadruple per step. If the parent-distance sources pay a fraction s_P of the child's compression, (3) gives $s_P I_{k+1} \lesssim c \Omega_k$, where Ω_k is the oscillation of the parent's vorticity and c is an absolute constant times the number of octaves. This is bounded amplification per step, the shape of Tao's averaged construction [27]. It is not enough by itself: whether the Biot–Savart transfer from parent to child, together with the self-supported share, can reach a factor 4 per halving at the viscous edge is a question about the constants in the ledger. In the simulations of [27] the realized factor is about $2^{0.4} \approx 1.3$.

Question 2.12 (transfer constant). *What is the largest per-halving intensity ratio that the exact Biot–Savart transfer, from parent-distance sources and from the event's own structure, can sustain for events at the viscous edge? Is it below 4?*

A value below 4 would answer Question 2.11 in the form that matters, since chains could not stay at the edge. A value at or above 4 would identify the configuration that a blowup must realize.

3. EVENTS IN FREQUENCY SPACE

We use the sharp dyadic bands of Yoneda, Goto and Tsuruhashi [24]: for $k \geq 0$, $f_k = \mathcal{F}^{-1}(\chi_{\{2^k/\sqrt{2} \leq |\xi| < \sqrt{2}2^k\}} \hat{f})$, applied to u and ω . Norms are on $\mathbb{T}^3$ with $|\mathbb{T}^3| = (2\pi)^3$.

Definition 3.1. The *band intensity* is $I_k = \|\nabla u_k\|_{L^\infty}$, the *band Reynolds number* is $R_k = I_k 2^{-2k}/\nu$, and the *escalation* is $\alpha_k = \log_2(I_{k+1}/I_k)$. The *band concentrations* are

$$\varphi_k^S = \frac{\|\nabla u_k\|_{L^2}^2}{|\mathbb{T}^3| \|\nabla u_k\|_{L^\infty}^2}, \quad \varphi_k^\omega = \frac{\|\omega_k\|_{L^2}^2}{|\mathbb{T}^3| \|\omega_k\|_{L^\infty}^2},$$

the effective fraction of the domain occupied by band k . If $\varphi_k \sim 2^{-(3-d)k}$, the band concentrates on a set of effective dimension d : $d = 3$ is spread out, $d = 2$ sheets, $d = 1$ filaments, $d = 0$ points.

Since the scale halves exactly from band to band,

$$\log_2 \frac{R_{k+1}}{R_k} = \alpha_k - 2, \quad (4)$$

the bookkeeping identity of Part III with $w_k = \log 2$; the critical escalation is again $\alpha = 2$.

Lemma 3.2 (what is allowed a priori). $\varphi_k^S, \varphi_k^\omega \geq c 2^{-3k}$ with an absolute constant c .

Proof. A band contains at most $N_k \leq C 2^{3k}$ Fourier modes, so by Cauchy–Schwarz and Parseval $\|f_k\|_{L^\infty}^2 \leq N_k \|f_k\|_{L^2}^2 / |\mathbb{T}^3|$. $\square$

A band can concentrate down to a single cell of its own size: dimension 0.

Proposition 3.3 (what the viscous edge requires). *If $R_k \geq R_c$, then $\varphi_k^S \leq \frac{4E_0}{|\mathbb{T}^3| R_c^2 \nu^2} 2^{-2k}$, where $E_0 = \frac{1}{2} \|u_0\|_{L^2}^2$.*

Proof. $\|\nabla u_k\|_{L^2} \leq \sqrt{2} 2^k \|u\|_{L^2} \leq 2^{k+1} E_0^{1/2}$, so

$$I_k = \|\nabla u_k\|_{L^2} (|\mathbb{T}^3| \varphi_k^S)^{-1/2} \leq 2^{k+1} E_0^{1/2} (|\mathbb{T}^3| \varphi_k^S)^{-1/2},$$

and $R_k \geq R_c$ gives the claim. $\square$

With only the energy, a band can stay at the viscous edge only if it concentrates at least as strongly as a filament: $d \leq 1$.

Lemma 3.4 (triad efficiency is bounded by concentration). *For the normalized triad of [24],*

$$\tilde{P}(2^k, 2^m, 2^\ell) = \frac{\int \omega_\ell \cdot \nabla u_m \cdot \omega_k}{\|\omega_\ell\|_{L^2} \|\omega_m\|_{L^2} \|\omega_k\|_{L^2}} \leq (|\mathbb{T}^3| \varphi_k^\omega)^{-1/2} \quad \text{for all } m, \ell.$$

Proof. Place ω_k in L^∞ and use $\|\nabla u_m\|_{L^2} = \|\omega_m\|_{L^2}$ for divergence-free fields. $\square$

Corollary 3.5 (regularity above dimension one). *If there are $\beta < 1$ and $c > 0$ with $\varphi_k^\omega(t) \geq c 2^{-2\beta k}$ for all k and all $t < t_*$, then the solution extends beyond t_* .*

Proof. By Lemma 3.4, the hypothesis of [24, Thm. 1] holds with exponent β . $\square$

In the language of dimension, $\varphi_k \geq c 2^{-2\beta k}$ means $d \geq 3 - 2\beta$: the criterion $\beta < 1$ is concentration of dimension strictly above one, and the trivial exponent $\beta = 3/2$ of Lemma 3.2 is dimension zero.

A measurement. For orientation only, an exploratory computation (not registered) on the saved 128^3 fields from the simulations of [27, §4] gives local effective dimensions $d = 3 - \log_2(\varphi_j^\omega / \varphi_{j+1}^\omega)$ between 1.5 and 2.5 for the random-phase flow, decreasing toward small scales, and between 0.6 and 3.5 for the Taylor–Green flow, whose coherent symmetric structures produce isolated octaves at or below one. Typical states sit above the critical dimension, with excursions toward it, consistent with the finding of [24] that their DNS satisfies $\beta < 1$. The top band is partly cut by the dealiasing.

4. COMPARISON

The two formalizations agree.

- (1) Both carry the same bookkeeping identity, (2) and (4), with the critical escalation $\alpha = 2$ equivalent to a constant event Reynolds number.
- (2) In frequency space, at the level of the a priori information, a band at the viscous edge must concentrate with dimension $d \leq 1$ (Proposition 3.3), and concentration with dimension $d > 1$, uniformly with a margin, gives regularity (Corollary 3.5). With band amplitudes saturating the energy bound, $d = 1$ is exactly $\alpha = 2$. So the critical lines coincide: $\alpha = 2$ in the event picture, $\beta = 1$ for the triad efficiency, $d = 1$ for the concentration.

- (3) The gap between what is known and what is needed is the same half derivative in every language: $\beta = 3/2$ against $\beta = 1$, dimension 0 against dimension 1.

The object of Question 2.11 is therefore not an artifact of how events are cut out of a flow. Two independent definitions, one following particles and one following Fourier bands, lead to the same critical line, and it is the line of the classical scaling gap. In frequency language the transfer question becomes a question about concentration: can the vorticity of a Navier–Stokes solution sustain effective dimension at most one over infinitely many octaves, with the parent-to-child Biot–Savart transfer supplying the factor 4 per octave that the edge requires?

Two classical results frame this. Caffarelli, Kohn and Nirenberg [4] show that the singular set of a suitable weak solution has vanishing one-dimensional parabolic Hausdorff measure: singularities, if any, live below dimension one in the parabolic sense, the analogue of the critical dimension here, which is where Proposition 3.3 places the edge. The sparseness programme of Grujić and collaborators [5] measures concentration directly, and its a priori content is exactly the energy [3], that is, the dimension-zero end of Lemma 3.2. What is missing is the same in all three languages: a mechanism, which by Parts I–III must use transport and the exact Biot–Savart law, that keeps active concentration above dimension one.

5. THE CURRENCY: CIRCULATION

For a vortex-like structure of scale ℓ and intensity $I \sim |\omega|$, the event Reynolds number is $R = I\ell^2/\nu \sim |\omega|\ell^2/\nu$, which is its circulation divided by ν . By (2), the critical escalation $\alpha = 2$ is a chain along which circulation is handed down intact. Circulation, unlike strain, is conserved by transport, and in the stochastic Lagrangian frame it is conserved exactly, with viscosity included.

Let $\tilde{u}$ be the random velocity of Constantin and Iyer, $\tilde{u} = \mathbb{P}[(\nabla^T A)(u_0 \circ A)]$, so that $u = \mathbb{E} \tilde{u}$ [1, (2.16)]. Restarting the representation at any time s with data $u(s)$, write $A_{s,t} = X_{s,t}^{-1}$.

Proposition 5.1 (noise-averaged Kelvin). *For every closed curve C and every $s \leq t < T^*$,*

$$\oint_C u(t) \cdot dr = \mathbb{E} \oint_{A_{s,t}(C)} u(s) \cdot dr,$$

and for every smooth oriented surface Σ ,

$$\int_{\Sigma} |\omega(t) \cdot n| dS \leq \mathbb{E} \int_{A_{s,t}(\Sigma)} |\omega(s) \cdot n| dS.$$

Proof. By [1, Prop. 2.9], $\oint_{X(\Gamma)} \tilde{u} \cdot dr = \oint_{\Gamma} u_0 \cdot dr$ for every closed curve Γ and every realization; take $\Gamma = A(C)$ and average, using $u = \mathbb{E} \tilde{u}$. Applied to the boundaries of all subsurfaces, the same identity says that the flux of $\text{curl } \tilde{u}$ through $X(\Gamma)$ equals the flux of ω_0 through Γ for every subsurface, so the total variation of the flux is transported pathwise; averaging and $|\mathbb{E} \cdot| \leq \mathbb{E} |\cdot|$ give the inequality. $\square$

Circulation is therefore never created by the dynamics. It is carried, compressed, and lost only through the noise average, that is, through viscosity. Two consequences follow. For $L > 0$ let $\Gamma_s^{\text{cap}}(L) = \sup\{|\oint_C u(s) \cdot dr| : \text{length}(C) \leq 2\pi L\}$ be the circulation capacity of $u(s)$ at scale L .

Corollary 5.2 (circulation is borrowed from larger scales). $\Gamma_t^{\text{cap}}(\ell) \leq \Gamma_s^{\text{cap}}(\ell e^{\Lambda_{s,t}})$, where $\Lambda_{s,t} = \int_s^t \|\nabla u\|_{L^\infty}$.

Proof. $\text{length}(A_{s,t}(C)) \leq \text{Lip}(A_{s,t}) \text{length}(C) \leq e^{\Lambda_{s,t}} \text{length}(C)$ for every realization; apply Proposition 5.1. $\square$

A structure carrying circulation Γ at scale ℓ must have been assembled by compressing circulation that was present at a larger scale; the compression is paid for in units of $\int \|\nabla u\|_{L^\infty}$, the currency of the levels of §2.

The known constructions in this currency. In the language of Proposition 5.1, the Navier–Stokes ledger has two properties together: circulation is transported exactly (a Kelvin theorem, pathwise in the stochastic frame), and it is averaged (the noise). Every blowup or growth construction in Table 1 of [27] gives up one of the two. The Euler constructions (Kiselev–Šverák, Elgindi, Chen–Hou, Tao’s Lagrangian modifications, and the claimed construction on $\mathbb{R}^3$) keep an exact Kelvin theorem and have no averaging. Tao’s averaged equation and Miller’s strain models keep viscosity but have no circulation theorem: the first because its nonlinearity is not the transport of vorticity, the second because it has no transport. No known construction carries circulation exactly *and* averages it.

A measurement. Exploratory, not registered: on the same saved fields, with $\bar{\omega}_\ell$ the vorticity filtered at scale ℓ and $R^\Gamma(\ell) = \pi\ell^2 \max |\bar{\omega}_\ell|/\nu$ the largest circulation Reynolds number at scale ℓ , the per-halving exponent $\alpha_\Gamma = 2 + \log_2(R^\Gamma(\ell/2)/R^\Gamma(\ell))$ is between 2.2 and 2.7 from $\ell = 0.8$ to 0.4 in the random-phase flow, where R^Γ is 100–260. It falls to 1.4–1.7, then 0.9–1.3, then below 1 as R^Γ drops to about 20–60. The Taylor–Green flow loses circulation from larger R^Γ . Circulation is carried essentially intact while its Reynolds number is large and is lost progressively as it approaches a few tens. That is the empirical shape of the threshold of Question 2.11, in its natural currency.

Question 5.3 (circulation transport). *Is there R^* such that, for the Navier–Stokes equations, the noise average of Proposition 5.1 removes a fixed fraction of the circulation handed from a structure to a structure at half its scale whenever the circulation Reynolds number is below R^* ?*

A positive answer is a viscous threshold in the sense of Question 2.11, and it would be a statement about the only mechanism that Navier–Stokes has and every known blowup construction lacks. It asks for a quantitative form of the averaging in Proposition 5.1: the preimages $A_{s,t}(C)$ of a small curve under different realizations of the noise are displaced from one another by about $\sqrt{\nu(t-s)}$, which at a circulation Reynolds number of order one is comparable to the curve itself, so averaging over them cancels circulation that is not coherent over that displacement. Making this quantitative requires control of the coherence of the vorticity over the diffusive displacement, which is again the self-consistency of the ledger.

6. A PARTIAL ANSWER: THE BURGERS CLASS

We settle Question 5.3 in a class of exact solutions: vortex tubes stretched by an axisymmetric strain, the setting of the Burgers vortex [6].

Lemma 6.1 (time-dependent Burgers vortex). *In the strain $u_s = (-\frac{\gamma}{2}x, -\frac{\gamma}{2}y, \gamma z)$, $\gamma > 0$, the vorticity $\omega = \frac{\Gamma}{\pi a(t)^2} e^{-r^2/a(t)^2} e_z$ is an exact solution of the Navier–Stokes equations provided $\frac{d}{dt}a^2 = 4\nu - \gamma a^2$, that is $a(t)^2 = a_B^2 + (a_0^2 - a_B^2)e^{-\gamma t}$ with $a_B^2 = 4\nu/\gamma$. The circulation Γ is constant, and the circulation inside radius ρ is $\Gamma(1 - e^{-\rho^2/a^2})$.*

Proof. With $\xi = r^2/a^2$, the three terms of $\partial_t \omega = \frac{\gamma}{2}r\partial_r \omega + \gamma\omega + \nu\Delta\omega$ are $\omega(a^2)'(\xi - 1)/a^2$, $-\gamma\xi\omega + \gamma\omega$ and $4\nu(\xi - 1)\omega/a^2$, and they balance exactly when $(a^2)' = 4\nu - \gamma a^2$. The induced swirl is axisymmetric and does not interact with the axisymmetric vorticity. $\square$

Proposition 6.2 (Burgers transfer bound). *In the setting of Lemma 6.1, for every $t \geq 0$ the circulation inside half the initial radius satisfies*

$$\Gamma(1 - e^{-a_0^2/(4a(t)^2)}) \leq \Gamma\left(1 - \exp\left(-\max\left\{\frac{1}{4}, \frac{\gamma a_0^2}{16\nu}\right\}\right)\right).$$

The strain can compress the tube to half its initial radius only if $\gamma a_0^2 > 16\nu$, and even then it never delivers the full circulation into that radius.

Proof. By Lemma 6.1, $a(t) \geq \min\{a_0, a_B\}$ for all t , and the flux inside a fixed radius increases as a decreases. $\square$

Corollary 6.3 (a parent line vortex). *Let the strain be supplied by a parent vortex of circulation Γ_p , modelled as a Lamb–Oseen vortex, at distance $d \geq a_0$ from the axis of the stretched tube. Then $\gamma \leq \Gamma_p/(2\pi d^2)$, so the fraction of the stretched tube’s circulation delivered inside half its initial radius is at most*

$$f(R_p) = 1 - e^{-R_p/(32\pi)}, \quad R_p = \Gamma_p/\nu.$$

Proof. The strain of a Lamb–Oseen vortex of circulation Γ_p and core ℓ at radius r has principal rates $\pm \frac{\Gamma_p}{2\pi r^2}(1 - (1 + r^2/\ell^2)e^{-r^2/\ell^2})$, bounded by $\Gamma_p/(2\pi r^2)$; so $\gamma a_0^2/(16\nu) \leq \Gamma_p a_0^2/(32\pi \nu d^2) \leq R_p/(32\pi)$. $\square$

Corollary 6.4 (chains in the Burgers class die). *Consider a hand-off chain in which, at each step, a parent of circulation Reynolds number R_k stretches a seed of circulation at most its own, in the configuration of Corollary 6.3, and the child is the part of the seed within half its initial radius. Then $R_{k+1} \leq R_k(1 - e^{-R_k/(32\pi)})$. Hence R_k decreases strictly, $R_k \rightarrow 0$, and by (2) the mean escalation is strictly subcritical: $\alpha_k - 2 = \log_2(R_{k+1}/R_k) \leq \log_2(1 - e^{-R_k/(32\pi)}) < 0$.*

Proof. The first inequality is Corollary 6.3. The sequence is decreasing and bounded below, and a positive limit L would satisfy $L = L(1 - e^{-L/(32\pi)})$, which is impossible. $\square$

In this class Question 5.3 has a positive answer for every R^* : the fraction removed per halving is at least $e^{-R/(32\pi)}$. At large circulation Reynolds number the loss per step is exponentially small and a chain can run for many octaves, of order $e^{R_0/(32\pi)}$ steps from R_0 up to algebraic factors, with finite total duration and enormous but finite intensity, before it drops below the threshold. That is transient growth, not a singularity; the model ignores the energy and geometric constraints that end real chains much earlier. The threshold $32\pi \approx 100$ is where the measurement of §5 sees circulation stop being carried intact. The model’s retained fraction, $\alpha_\Gamma = 2 + \log_2 f(R)$, gives $\alpha_\Gamma \approx 1.8$ at $R = 200$, 1.3 at $R = 100$ and 0.6 at $R = 50$. This is the trend of the measured exponents, in which the largest-scale values above 2 reflect the maximum over several structures rather than a single chain.

The limits of the class must be stated. The strain is axisymmetric and uniform, whereas the strain of a parent line vortex is planar; for Burgers vortices in non-axisymmetric strain at large circulation Reynolds number the core is, to leading order, still set by the axial rate [15]. The parent is a single line vortex; several parents can supply more strain, bounded in general only by the shell estimate (3). And the seed is assumed to carry at most the parent’s circulation; Proposition 5.1 forbids creating circulation but not drawing a stronger seed from elsewhere. Within these limits, the averaging mechanism of §5 is quantitative: a strain cannot compress circulation below its Burgers radius, and the Burgers radius is the viscous threshold.

A numerical check, and its limits. A registered test on the tracer data of [27, §4] did not confirm the dependence of retention on R predicted by Corollary 6.3. It used the circulation proxy $R_j^\Gamma = \pi \sigma_j^2 |\bar{\omega}_j|/\nu$ at the filter widths σ_j . The median retention per halving was about

0.25–0.43 and nearly flat over $R = 10$ –160, and it never exceeded one, as Proposition 5.1 requires. On reading the results we found the proxy inadequate: a structure that merely persists, seen at half the filter width, gives retention 1/4 with no hand-off, so the test mostly measured how the filtered vorticity scales at a tracer. The Burgers threshold is therefore neither confirmed nor refuted by this test. A proper test must track the circulation of identified vortical structures.

7. BEYOND THE BURGERS CLASS

The three limits of §6 were the Gaussian profile and axisymmetric strain, the single parent, and the seed's strength. All three can be removed in the exact reduction that underlies the Burgers vortex and its asymmetric versions [15, 7]: a vortex tube uniform along its axis in an external linear strain,

$$u = (D(t)x + u_\omega(x, t), s(t)z), \quad x \in \mathbb{R}^2, \quad \text{tr } D = -s, \quad (5)$$

with $D(t)$ symmetric and otherwise arbitrary (time-dependent, non-axisymmetric), and u_ω the planar velocity induced by the axial vorticity $\omega(x, t)$ through the two-dimensional Biot–Savart law. This is an exact solution of the Navier–Stokes equations whenever ω solves

$$\partial_t \omega + \nabla \cdot ((Dx + u_\omega)\omega) = \nu \Delta \omega, \quad (6)$$

where the divergence of Dx , equal to $-s$, cancels the stretching term $s\omega$.

Theorem 7.1 (a structure cannot be compressed below its strain radius). *Let $\omega \geq 0$ solve (6), with $\Gamma = \int \omega$ and finite second moment. Let $x_c = \Gamma^{-1} \int x\omega$ be its centroid and $r_c^2 = \Gamma^{-1} \int |x - x_c|^2 \omega$ its mean-square radius, and let $\gamma(t) \geq 0$ be the largest compression rate of $D(t)$, that is $-\gamma$ is its smallest eigenvalue. Then Γ is constant, $x'_c = Dx_c$, and*

$$\frac{d}{dt} r_c^2 \geq -2\gamma r_c^2 + 4\nu, \quad \text{hence} \quad r_c(t)^2 \geq \min \left\{ r_c(0)^2, \frac{2\nu}{\sup_{[0,t]} \gamma} \right\}.$$

The bound is attained by the Burgers vortex.

Proof. From (6), $\frac{d}{dt} \int \varphi \omega = \int \omega (Dx + u_\omega) \cdot \nabla \varphi + \nu \int \omega \Delta \varphi$ for smooth φ . With $\varphi = 1$, Γ is constant. With $\varphi = x$, $\frac{d}{dt} (\Gamma x_c) = D \Gamma x_c + \int \omega u_\omega = D \Gamma x_c$, because $\int \omega u_\omega = \iint \omega(x) \omega(y) K(x - y) dx dy = 0$ for the odd kernel $K(z) = z^\perp / (2\pi |z|^2)$. With $\varphi = |x - x_c|^2$, and using $\int \omega(x - x_c) = 0$,

$$\frac{d}{dt} \int |x - x_c|^2 \omega = 2 \int \omega (x - x_c) \cdot D(x - x_c) + 2 \int \omega (x - x_c) \cdot u_\omega + 4\nu \Gamma.$$

The middle term vanishes. Symmetrizing, $\iint \omega(x) \omega(y) (x - y) \cdot K(x - y) = 0$ because $z \cdot z^\perp = 0$, and the x_c part vanishes as before. This is the conservation of angular impulse by the planar self-induced flow. Since $\omega \geq 0$, the first term is at least $-2\gamma \int \omega |x - x_c|^2$. Dividing by Γ gives the differential inequality, and comparison gives the bound. For the Burgers vortex of Lemma 6.1 the in-plane compression rate is $\gamma/2$ and $r_c^2 = a^2$, and the steady radius $a_B^2 = 4\nu/\gamma$ equals $2\nu/(\gamma/2)$. $\square$

For the planar Navier–Stokes equations without strain, the growth of the second moment at the rate $4\nu\Gamma$ and the vanishing of the self-induced term are classical; they are used, for instance, in the stability theory of Burgers vortices [7]. What is new here is only the use of the moment with an external strain to obtain a threshold. Three consequences remove the three limits.

Any profile, any planar strain. Theorem 7.1 needs neither a Gaussian profile nor axisymmetry nor steadiness of the strain. Measured by its mean-square radius, a tube can be compressed to half its size only if

$$\sup \gamma \geq 8\nu/r_c(0)^2. \quad (7)$$

A structure cannot compress itself. The self-induced flow contributes exactly nothing to $\frac{d}{dt}r_c^2$. However strong the seed, only external strain compresses it. This removes the limitation on the seed's strength: the threshold is on the stretchers, not on the structure being stretched.

Several parents. The external strain at the tube is the sum of the strains of all the other structures; to the extent that it is nearly uniform across the tube, which holds when the other structures are farther away than the tube's size, Theorem 7.1 applies with it. For line vortices of circulations Γ_i at distances d_i , $\gamma \leq \sum_i \Gamma_i / (2\pi d_i^2)$; in general it is bounded by the shell estimate (3). With $d_i \geq r_c(0)$, (7) requires

$$\sum_i R_i \frac{r_c(0)^2}{d_i^2} \geq 16\pi, \quad R_i = \Gamma_i / \nu.$$

The stretchers of a halving must together carry a circulation Reynolds number of at least $16\pi \approx 50$ at distances comparable to the structure's size. In this class this is the viscous threshold of Question 2.11, with $R_c = 16\pi$, now on the *total* circulation of the stretchers.

A test on identified tubes. In a registered test, we ran new simulations at 128^3 with $\nu = 1/200$, where the resolution satisfies $k_{\max}\eta \geq 2$ at every analysed time. We identified intense vortex tubes as connected components of $|\omega| > 3\omega_{\text{rms}}$ with elongation at least 2. For each tube we split the strain exactly into the part induced by the tube's own vorticity and the rest, with the Biot–Savart law applied to masked fields. The instrument recovered the radius of a synthetic Gaussian tube to 2%, gave zero self-compression for an isolated straight tube, and gave the compression by an antiparallel partner within 4% of $\Gamma / (2\pi d^2)$. With γ_{ext} the in-plane compression rate of the external strain, the ratio $B = r_c^2 \gamma_{\text{ext}} / (2\nu)$, which Theorem 7.1 predicts to be at least 1 at balance, had median 1.01, weighted by $\int |\omega|$, in the random-phase flow (105 tubes) and 1.21 in the Taylor–Green flow (128 tubes, about four independent by symmetry); $B \geq 1/2$ held for 91% and 100% of the tubes. Real tubes sit at the strain radius of Theorem 7.1, set by the *external* strain. This is consistent with, and in its first half replicates, the finding of Jiménez and Wray [20] that intense filaments are inhomogeneous Burgers vortices driven by the strain of the background. The statement that a structure cannot compress itself held in the random-phase flow (the median ratio of self- to external compression was 0.085) but only marginally in the Taylor–Green flow (0.54, whose tubes are strongly curved and interacting). That is where the reduction is expected to fail: in three dimensions a curved structure does strain itself.

What is left: branching. Theorem 7.1 and Proposition 5.1 together reduce the continuation of a chain to an accounting of stretchers. Each halving needs stretchers with total circulation at least $16\pi\nu$ nearby. Circulation is never created, only compressed from larger scales, and a structure cannot compress itself. Whether a chain can go on indefinitely therefore depends on how stretchers are reused. If every halving of every structure required a new stretcher, the stretchers would form a tree that doubles at each level. Estimating, only heuristically, the energy of a structure of circulation Γ and size ℓ as of order $\Gamma^2\ell$, the energy of such a tree grows linearly with its depth, which would then be bounded by a multiple of $E_0 / ((16\pi\nu)^2 \ell_0)$. A chain in which one stretcher serves successive generations costs only a geometric series of energy and is not bounded this way; the claimed Euler construction [8] is organized exactly so, with seeds placed in advance.

Question 7.2 (branching). *Must the stretchers of successive halvings in a Navier–Stokes solution be distinct, so that the circulation budget branches, or can one structure of circulation above $16\pi\nu$ drive the compression of arbitrarily many successive generations?*

This is where the transfer question now stands. It is a question about the reuse of stretchers, and it is a question that the viscous threshold, now proved in the reduction, has made precise.

A first measurement, and what it changed. In a registered test on new simulations (128^3 , $\nu = 1/200$) we tracked the thirty most intense structures every 0.1 time units. For each pair we computed the part of one structure's compression supplied by the other, with the Biot–Savart law applied to masked fields. By the registered rule, stretchers were reused: the median stretcher compressed two distinct structures over its life, in both flows. But the same computation showed that 82% (Taylor–Green) and 90% (random phases) of each tube's compression came from vorticity outside all thirty structures, and that no single structure ever compressed a tube faster than viscous spreading. At these Reynolds numbers the stretcher is not a distinct structure but the collective field, shared by every structure it acts on, which is reuse in the extreme. The branching question as posed presumes discrete stretchers; in these flows it returns to the collective strain at each scale, that is, to the frequency-local formulation of §3. Whether discrete stretchers emerge at higher Reynolds number, as the hierarchy of Goto and collaborators suggests, is open.

8. CURVATURE: ONLY AXIAL STRETCHING COMPRESSES

In three dimensions a curved structure strains itself, and the Taylor–Green tubes of §7 showed it. The question is whether that self-strain can compress a structure below its strain radius. In the reduction (5) it cannot, and the reason is that only the axial stretching s compresses. For $\omega \geq 0$ on $\mathbb{R}^2$ let

$$K_t(\rho) = \sup \left\{ \int_A \omega(x, t) dx : |A| = \pi \rho^2 \right\}$$

be the concentration function, the largest circulation that fits in any set of the area of a disk of radius ρ .

Theorem 8.1 (only axial stretching compresses). *Let $\omega \geq 0$ solve (6) with $D = -\frac{s}{2}I + D_0$, $\text{tr } D_0 = 0$, and let $S(t) = \int_0^t s$. Then for all $\rho > 0$ and $t \geq 0$,*

$$K_t(\rho) \leq K_0(\rho e^{S(t)/2}).$$

The trace-free part D_0 of the strain and the self-induced flow compress nothing; the circulation that can be packed into radius ρ at time t is at most what was initially within the radius $\rho e^{S/2}$ from which axial stretching can have compressed it.

Proof. Set $\xi = e^{S/2}x$ and $\omega = e^S \Omega(\xi, t)$, which preserves circulation. The dilation $-\frac{s}{2}x$ is absorbed by the change of variables, the two-dimensional Biot–Savart law is covariant under it, and a direct computation turns (6) into

$$\partial_t \Omega + \nabla_\xi \cdot (V \Omega) = \nu e^S \Delta_\xi \Omega, \quad V = D_0 \xi + e^S \tilde{u}_\Omega, \quad \nabla_\xi \cdot V = 0,$$

where $\tilde{u}_\Omega$ is the velocity induced by Ω . By the bathtub principle, the concentration function of Ω is $K_t^\Omega(r) = \sup \{ \int \varphi \Omega(t) : 0 \leq \varphi \leq 1, \int \varphi = \pi r^2 \}$. Given such a φ and $\tau < t$, let $\tilde{\varphi}$ solve the adjoint problem $-\partial_\sigma \tilde{\varphi} - V \cdot \nabla \tilde{\varphi} = \nu e^S \Delta \tilde{\varphi}$ on $[\tau, t]$ with $\tilde{\varphi}(t) = \varphi$. Then $\int \varphi \Omega(t) = \int \tilde{\varphi}(\tau) \Omega(\tau)$. By the maximum principle $0 \leq \tilde{\varphi} \leq 1$, and since V is divergence free $\int \tilde{\varphi}$ is conserved. So $\tilde{\varphi}(\tau)$ is admissible and $K_t^\Omega \leq K_\tau^\Omega$. Finally $K_t(\rho) = K_t^\Omega(\rho e^{S/2}) \leq K_0^\Omega(\rho e^{S/2}) = K_0(\rho e^{S/2})$. $\square$

For bounded divergence-free drifts on $\mathbb{R}^d$, Hess–Childs, Raquépas and Rowan prove the stronger statement that such drifts decrease concentration relative to the heat equation [19]; the linear strain here is unbounded, which is why we use the weaker, elementary monotonicity. Together with Theorem 7.1, which gives the viscous floor, the theorem says that compressing circulation requires axial stretching and cannot go below the strain radius.

Curvature. For a slender filament, the leading self-induced motion is the local induction, or binormal, flow, which preserves arc length [18]. It produces no axial self-stretching, so by Theorem 8.1 a slender curved filament does not compress itself at leading order; the trace-free strain that its curvature induces on its core only deforms it. Self-compression requires axial self-stretching, which in three dimensions comes from nonlocal self-interaction: distant parts of the same structure acting on each other, as in hairpins and antiparallel approaches. At the level of a segment those parts are external stretchers, and the threshold of §7 applies to them.

A test. A registered test on the snapshots of §7 split each tube into axial segments of length four times its radius and recomputed the self- and external compression for each segment. In the Taylor–Green flow the fraction of tubes in which the tube’s own vorticity compressed it less than the external vorticity rose from 0.72 at the level of whole components to 1.00 at the level of segments. The median ratio of self- to external compression fell from 0.54 to 0.23. In the random-phase flow, whose tubes are less curved, the ratio did not fall (0.085 to 0.113), so the drop in the Taylor–Green flow is not an artefact of splitting. The self-compression of curved tubes at component level is the stretching of one part of a structure by another part.

9. WHERE THE COLLECTIVE QUESTION LANDS

§7 ended with the observation that at moderate Reynolds number structures are compressed by the collective field rather than by distinct stretchers, so the transfer question is the band question of §3: can the collective strain stay at the viscous edge across infinitely many octaves? Two elementary accountings show where this question sits.

Proposition 9.1 (event dissipation budget). *Suppose that for each band $k \geq k_0$ there is a time interval of length at least c/I_k during which $|\nabla u_k| \geq I_k$ on a set of measure at least m_k , with $I_k \geq R_c \nu 4^k$. Then*

$$\sum_{k \geq k_0} 4^k m_k \leq \frac{E_0}{c R_c \nu^2}.$$

In particular, if $m_k \geq c' 2^{-(3-d)k}$ then $d < 1$.

Proof. The bands are orthogonal, so by the energy inequality $\sum_k \nu \int \|\nabla u_k\|_{L^2}^2 dt \leq E_0$, and the hypothesis gives $\nu \int \|\nabla u_k\|_{L^2}^2 \geq \nu m_k I_k^2 c/I_k = c \nu m_k I_k \geq c R_c \nu^2 4^k m_k$. If $m_k \geq c' 2^{-(3-d)k}$, then $4^k m_k \geq c' 2^{(d-1)k}$, which is not summable when $d \geq 1$. $\square$

A chain held at the edge across infinitely many octaves must therefore live, at each scale, on a set of dimension strictly below one. Point-like collapse is allowed by the energy; a cascade of filaments is not.

The circulation picture. By Proposition 5.1 circulation is carried and never created, and by Theorem 8.1 only axial stretching compresses. A critical chain, one along which the circulation Reynolds number stays at the edge, is then one circulation of at least $16\pi\nu$ compressed through every scale by collective axial stretching whose rate grows like r^{-2} . By the Biot–Savart law that stretching is supplied by vorticity at distances comparable to r , so the neighbourhood must intensify with the core. With the circulation Reynolds number bounded, the velocity at scale r is of order ν/r : the scale-invariant bound $|u| \lesssim |x|^{-1}$, the Leray scaling in space. It fixes the spatial profile of the collapse, not its rate in time; the rate is taken up in §10.

Both conclusions are the classical frontier. A set of dimension below one is where the theorem of Caffarelli, Kohn and Nirenberg places singularities [4], and a neighbourhood intensifying at the Leray scaling in space is the class of scenarios addressed by the exclusion of self-similar solutions [9, 11] and by the Liouville theory of bounded ancient solutions [13], to which every blowup, of Type I or Type II, rescales. The transport route of this series reaches the same place

as the classical theory, from the other side. What it adds is a mechanism-level description of the approach: circulation as the currency (Proposition 5.1), a viscous threshold on the stretchers (Theorem 7.1), compression only by axial stretching (Theorem 8.1), and the measurements that locate real flows relative to each.

10. THE ANATOMY OF A COLLAPSE

We apply the tools of this part to the scenario that §9 singles out: a core that collapses while circulation is carried into it. The first observation is that the reduction does not exclude it, at any rate.

Proposition 10.1 (collapse at any rate). *In the setting of Lemma 6.1, let $a(t) > 0$ be smooth and nonincreasing on $[0, T)$ with $a(t) \rightarrow 0$ as $t \rightarrow T$, and let the axial strain be*

$$\gamma(t) = \frac{4\nu - (a^2)'(t)}{a(t)^2}.$$

Then the Gaussian vortex of radius $a(t)$ is an exact solution. It keeps its circulation Γ , its peak vorticity is $\Gamma/(\pi a^2)$, and its peak swirl velocity is of order Γ/a . In particular:

- (i) $a^2 = 4\nu(T - t)/(k - 1)$ with $k > 1$ gives $\gamma = k/(T - t)$; the radius shrinks like $(T - t)^{1/2}$ and the peak vorticity grows like $(T - t)^{-1}$, the Type I rate;
- (ii) $a^2 = c(T - t)^\beta$ with $c > 0$ and $\beta > 1$ gives $\gamma = 4\nu/(c(T - t)^\beta) + \beta/(T - t)$; the peak vorticity grows like $(T - t)^{-\beta}$ and the swirl velocity like $(T - t)^{-\beta/2}$, a Type II rate.

Proof. The condition of Lemma 6.1 is $(a^2)' = 4\nu - \gamma a^2$, which is the definition of γ , and $\gamma > 0$ because $(a^2)' \leq 0$. In (i), $(a^2)' = -4\nu/(k - 1)$, so $\gamma a^2 = 4\nu k/(k - 1)$. In (ii), $(a^2)' = -c\beta(T - t)^{\beta-1}$. $\square$

The collapse respects every statement proved above: circulation is conserved (Proposition 5.1); the radius stays above the strain radius, since $\gamma a^2 = 4\nu - (a^2)' \geq 4\nu$, that is $a^2 \geq 2\nu/(\gamma/2)$ (Theorem 7.1); and it is driven entirely by axial stretching (Theorem 8.1). In the reduction the strain is prescribed from outside, so none of these statements can exclude collapse, at any rate. An exclusion must use the fact that in a Navier–Stokes solution the strain is produced by the flow itself.

Corollary 10.2 (who supplies the strain). *If the axial strain at the collapsing core of Proposition 10.1 is supplied by vortices of circulations Γ_i at distances $d_i \geq a$ from its axis, then*

$$\sum_i \frac{\Gamma_i}{\nu} \frac{a^2}{d_i^2} \geq \frac{2\pi\gamma a^2}{\nu} = 8\pi \left(1 - \frac{(a^2)'}{4\nu}\right) \geq 8\pi.$$

The bound holds at every rate. In case (i) the middle term is $8\pi k/(k - 1)$; in case (ii) it tends to 8π .

Proof. The strain of a vortex of circulation Γ_i at distance d_i is at most $\Gamma_i/(2\pi d_i^2)$ (Corollary 6.3), so $\gamma \leq \sum_i \Gamma_i/(2\pi d_i^2)$. Multiply by $2\pi a^2/\nu$ and use $\gamma a^2 = 4\nu - (a^2)'$. $\square$

For a general profile in a general planar strain, Theorem 7.1 gives the same statement for the mean-square radius with 4π in place of 8π : a shrinking r_c needs a compression rate of at least $2\nu/r_c^2$, and each supplier contributes at most $\Gamma_i/(2\pi d_i^2)$.

The bound is weakest in the adiabatic limit $(a^2)'/\nu \rightarrow 0$, where the core sits at its steady Burgers radius. The suppliers must carry a total circulation Reynolds number of at least $8\pi \approx 25$ at distances comparable to the core, so, as long as their circulations stay bounded, they must intensify with it. If they are themselves collapsing tubes, each needs its own suppliers under the

same condition. A collapse at bounded circulation Reynolds numbers is thus a cluster of vortices that stretch one another axially, each carrying a circulation Reynolds number of order 10 to 100 at mutual distances comparable to their radii; symmetric configurations of this kind are the ones studied numerically as candidates for collapse [10]. By Proposition 5.1 the circulations of the members are conserved and only averaged, so the cluster cannot feed itself circulation; it must be assembled with its circulation in place.

Type I and Type II. For a tube the velocity at the core is of order $\Gamma/(2\pi a)$, so, heuristically,

$$\frac{\|u\|_{L^\infty} \sqrt{T-t}}{\sqrt{\nu}} \approx \frac{1}{2\pi} \cdot \frac{\Gamma}{\nu} \cdot \frac{\sqrt{\nu(T-t)}}{a}. \quad (8)$$

A collapse is of Type I if the left side stays bounded; case (i) of Proposition 10.1 keeps both factors on the right bounded, with the core at the parabolic scale $\sqrt{\nu(T-t)}$. A Type II collapse must make one of the two factors unbounded, and the two ways are different.

Remark 10.3 (two kinds of Type II collapse). (a) *Below the parabolic scale, with bounded circulation Reynolds number:* $a \ll \sqrt{\nu(T-t)}$, as in case (ii). The core's viscous time a^2/ν is then short compared with the time left, $(a^2)'/\nu \rightarrow 0$ along the collapse, the core sits at its steady Burgers radius, and Corollary 10.2 holds with its weakest constant. Write $u(x, t) = (\nu/a)U(x/a, s)$ with $ds/dt = \nu/a^2$. Then U solves the Navier–Stokes equations with $\nu = 1$ plus the term $-(aa'/\nu)(U + y \cdot \nabla U)$; the rescaled time runs to infinity and the coefficient aa'/ν tends to zero. A limit profile is therefore a bounded eternal solution of the Navier–Stokes equations themselves, not of an equation in similarity variables: a self-sustaining cluster of tubes that stretch one another at Burgers balance. This is the Liouville question of Koch, Nadirashvili, Seregin and Šverák for bounded ancient solutions [13], to which in fact every blowup rescales; the anatomy adds what a counterexample would have to contain. A heuristic link, not a statement: a steadily stretched tube is an axial jet (the Burgers vortex carries the axial velocity γz), so a steady limit profile would export fluid along its tubes over distances long compared with their radii; in the related stationary problem with finite Dirichlet integral, the capsule analysis of Coiculescu and Yang shows that a nontrivial solution must contain arbitrarily elongated coherent jets [14].

(b) *Gathered circulation:* $\Gamma/\nu \rightarrow \infty$ along the collapse. Proposition 5.1 forbids creating circulation but not gathering it from larger loops, and Corollary 5.2 does not bound the gathering, since $\Lambda_{s,t} \rightarrow \infty$ at a singularity. With the heuristic energy $\Gamma^2 \ell$ of §7, the energy alone allows Γ up to about $(E_0/\ell)^{1/2}$ at scale ℓ : the half-derivative gap, in the currency of circulation. The supplier bound then relaxes: suppliers of circulation Reynolds number R_i may sit at distances of order $a\sqrt{R_i/8\pi}$, far from the core, so the cluster can be replaced by a hierarchy separated in scale, a supercritical chain with $\alpha > 2$ in the sense of [27]. The reduction (5) conserves Γ exactly and cannot see this branch. In axisymmetric flow the swirl circulation ru^θ obeys a maximum principle, which closes this branch for the swirl and is the scalar on which the axisymmetric Liouville theorems of [13] rest.

Where the collapse lands. In the similarity variables $x/\sqrt{T-t}$ and $-\log(T-t)$, a Type I collapse is a bounded trajectory of Leray's equation, which is the rescaled equation of Remark 10.3(a) with a constant nonzero coefficient aa'/ν , as in case (i). Self-similar solutions are its fixed points, excluded in the classes of [9, 11]; discretely self-similar solutions are its periodic orbits, which have at most one singular point and are excluded when the scaling factor is close to one [12]; general bounded trajectories are open. Case (a) of Type II is the Liouville question of [13], and case (b) is the half-derivative gap, where the tools of this series say least. The noise-averaged Kelvin theorem settles none of these questions; it gives the anatomy of each.

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